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A MAJOR PROJECT REPORT
ON

"LITMUS : A Multimodal Guardrail Framework for Jailbreak Detection in Vision-Language Models"

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LIST OF ABBREVIATIONS

ACL	: Annual Meeting of the Association for Computational Linguistics
AI	: Artificial Intelligence
BERT	: Bidirectional Encoder Representations from Transformers
CLIP	: Contrastive Language-Image Pre-training
DeBERTa	: Decoding-enhanced BERT with Disentangled Attention
MLP	: Multilayer Perceptron
NLP	: Natural Language Processing
OCR	: Optical Character Recognition
ViT	: Vision Transformer
VLM	: Vision-Language Model

CHAPTER 1

INTRODUCTION

1.1 Background

Vision-Language Models (VLMs) extend large language models into the visual domain by pairing a vision encoder with a language model, allowing a single system to reason over an image and a text prompt together. This has enabled applications such as visual question answering, image-based search, and multimodal assistants, but it has also introduced safety concerns beyond those of text-only systems.

Model safety is typically achieved through alignment techniques such as instruction tuning, RLHF, and DPO, which train a model to refuse harmful requests. This safety, however, is learned behavior rather than a fixed rule, forming a soft barrier that can be bypassed using inputs the model was not trained to handle, a bypass known as a *jailbreak*. To reduce this risk without retraining the underlying model, systems deploy *guardrails*: model-agnostic safety layers that inspect inputs, and sometimes outputs, independently of the model itself. Most existing guardrails, however, were built for text and cannot inspect image content.

Vision inputs open new avenues for jailbreak attacks. In *typographic injection*, harmful instructions are rendered as text inside an image, readable by the vision component but invisible to text-only filters. In *query-relevant pairing*, an individually harmless text prompt and image combine to produce harmful output only when interpreted together. In *continuous-space adversarial perturbation*, small, often imperceptible pixel-level changes shift the model's internal representations enough to suppress its refusal behavior. Because pixel values are continuous, images offer a far larger attack space than text, making VLMs considerably harder to secure.

1.2 Motivation

As VLMs move into real-world use in search, moderation, customer support, and assistive tools, the consequences of an undetected multimodal jailbreak grow accordingly, while existing guardrails, built mainly for text, remain blind to attacks carried through the image channel. This project is motivated by the need for a lightweight, model-agnostic guardrail that inspects both modalities together before a request reaches the downstream VLM, closing this gap without requiring changes to the underlying model.

1.3 Problem Statement

Vision-Language Models (VLMs) can be vulnerable to jailbreak attacks where harmful instructions are hidden in images or combined with seemingly harmless text. Existing text-based safety guardrails may fail to detect these attacks. Therefore, this project aims to develop a multimodal guardrail that analyzes both text and image inputs to detect and classify potential jailbreak attacks, improving the safety of vision-language systems.

1.4 Objectives

The key objectives of this project are:

1. To design and implement a lightweight multimodal guardrail that detects jailbreak attempts in combined text-and-image inputs before they reach the downstream language model.
2. To evaluate the guardrail against three major multimodal jailbreak failure modes and measure its effectiveness using suitable evaluation resources.

1.5 Scope and Delimitations

The scope of this project is to develop a lightweight multimodal guardrail that detects jailbreak attempts in combined text-and-image inputs before they reach a downstream language model. The system will focus on detecting typographic injection, query-relevant pairing, and adversarial image perturbation attacks and will classify inputs as safe or potentially malicious. The project is limited to text and image inputs and does not cover audio or video-based attacks. It focuses specifically on jailbreak detection rather than addressing all possible AI security threats, and the evaluation will be limited to selected datasets, attack types, and experimental conditions.

CHAPTER 2

LITERATURE REVIEW

The deployment of Vision-Language Models (VLMs) has introduced cross-modal capabilities alongside severe security risks from multimodal jailbreak attacks. This chapter reviews safety alignment limitations, multimodal attack taxonomies, existing defense paradigms, and the research gaps motivating this project.

2.1 Safety Alignment and Multimodal Jailbreak Taxonomies

Safety alignment methods such as Reinforcement Learning from Human Feedback (RLHF) and Direct Preference Optimization (DPO) steer model responses toward predefined safety guidelines [1, 2]. However, alignment forms a soft probabilistic barrier rather than a strict constraint, allowing adversaries to induce policy violations through jailbreak attacks [3, 4]. In VLMs, visual inputs expand the attack surface across three primary failure modes [2, 5]:

- **Typographic Injections**: Malicious textual instructions are rendered visually within an image, bypassing text-only safety filters while instructing the visual reasoning module [5, 6].
- **Query-Relevant Pairing**: Harmless text is paired with a visual prompt where neither is individually toxic, but their combined semantic synergy triggers harmful generation [5, 7].
- **Continuous-Space Adversarial Perturbations**: Gradient-optimized pixel perturbations disrupt internal safety representations in the vision encoder, evading refusal triggers [4, 8].

2.2 Existing Defense Paradigms and Guardrails

Existing defense frameworks can be classified into prompt-level text guardrails, internal state monitoring, and multimodal feature fusion filters:

- **Text-Centric Guardrails**: Frameworks such as *JailGuard* [3] and deep transformer classifiers [4] evaluate prompt mutations and text semantics. However, they are incapable of inspecting image contents or cross-modal context.

- **Internal Representation Defenses**: Methods like *HiddenDetect* [9] track transformer layer hidden states, Zhu et al. [8] apply null-space projection steering, and *JailDAM* [10] employs latent memory networks. While effective, they require white-box access and introduce heavy computational overhead, making them impractical for black-box API deployment.

- **Multimodal Fusion Guardrails**: Hua et al. [6] demonstrated that multi-branch representations combining raw image features, OCR text, and user text improve detection accuracy. Similarly, Liang et al. [7] showed that decoupled visual-textual encodings enhance generalization to unseen attacks.

2.3 Summary of Related Works and Research Gaps

Table 2.1 summarizes key existing literature in relation to the proposed system.

Table 2.1: Comparative summary of existing jailbreak detection approaches.

Framework	**Modality**	**Core Strategy**	**Key Limitations**
JailGuard [3]	Text	Mutation and response scoring	Blind to visual prompt injections
HiddenDetect [9]	Vision+Text	Layer hidden-state monitoring	Requires white-box access; high latency
Zhu et al. [8]	Vision+Text	Null-space activation steering	High inference cost; architecture dependent
Hua et al. [6]	Vision+Text	Multi-branch feature scoring	Lacks calibrated gating for input filtering
Proposed System	**Vision+Text**	**Tri-branch Encoder + Gated MLP**	**Lightweight pre-VLM input-level guardrail**

Identified Gaps and Justification: Conventional defenses either ignore visual vectors or depend on computationally expensive white-box activation monitoring. Furthermore, unimodal vision classifiers fail on typographic and cross-modal synergistic attacks. To address these gaps, the proposed work develops a lightweight, model-agnostic guardrail leveraging a tri-branch architecture to filter harmful inputs before reaching the downstream VLM.

CHAPTER 3

METHODOLOGY

3.1 System Components

The proposed system is a lightweight multimodal guardrail designed to detect jailbreak attempts in combined text-and-image inputs before they reach the protected Vision-Language Model (VLM). The methodology consists of several stages, including input routing, feature extraction, feature projection, feature fusion, classification, threshold-based decision making, and final interaction with the protected VLM.

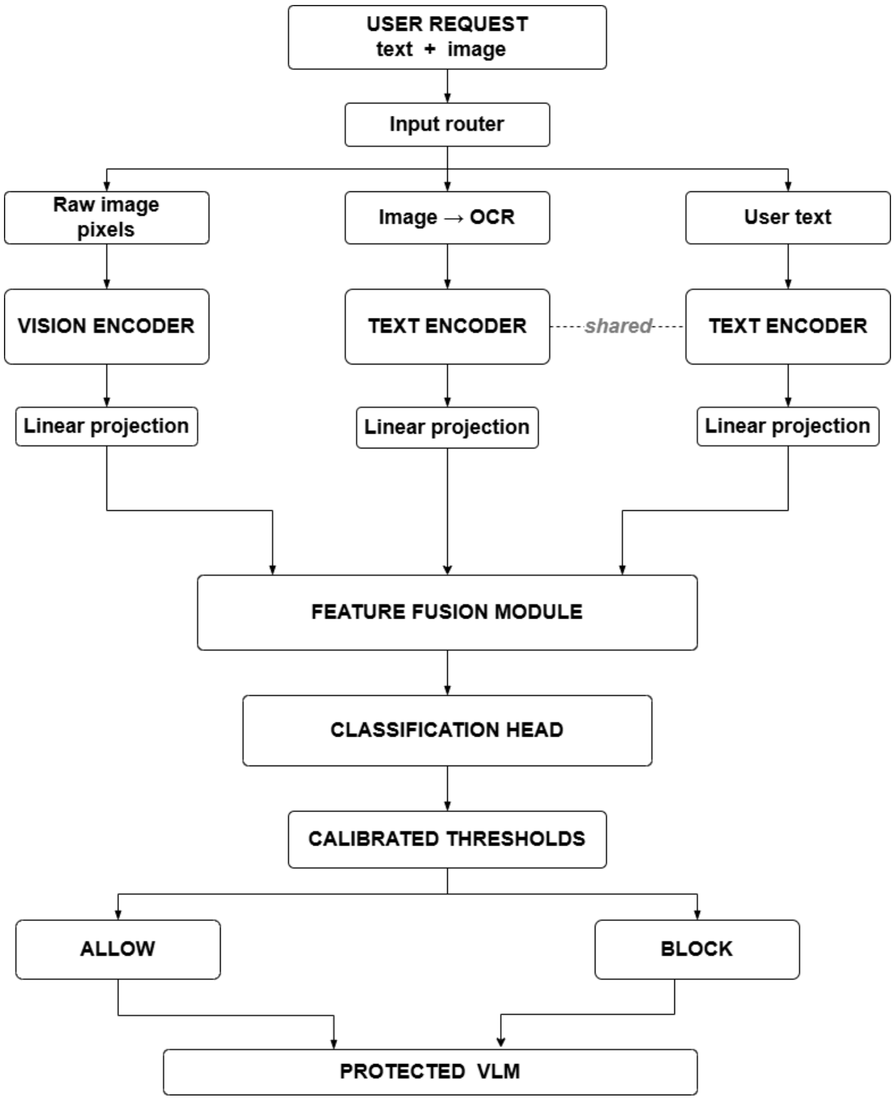

Figure 3.1: System Design Architecture for Multimodal Guardrail

3.1.1 User Request

The system first receives a user request containing text and an image. Both modalities are considered because a jailbreak instruction may be present directly in the user's text or hidden within the image.

3.1.2 Input Router

The input router receives the complete user request and separates it into three branches: raw image pixels, text extracted from the image using Optical Character Recognition (OCR), and the original user text. This separation allows the system to independently analyze the visual and textual information.

3.1.3 Raw Image Processing

The raw image pixels are directly passed to the vision encoder. This branch captures visual information from the image, including visual patterns, objects, layouts, and text-like features that may not be successfully extracted by OCR.

3.1.4 Vision Encoder

The raw image is processed using a vision encoder . The encoder converts the image into a numerical feature representation. These features contain information about the visual content of the input and are later used by the feature fusion module for jailbreak detection.

3.1.5 Image-to-OCR Processing

The image is also passed through an Optical Character Recognition (OCR) module to extract text present inside the image. This is important for detecting typographic injection attacks, where a malicious instruction is embedded as text within an image.

3.1.6 OCR Text Encoder

The text extracted by OCR is passed to a text encoder. The text encoder converts the extracted text into a numerical feature representation. These features provide semantic information about the text contained in the image.

3.1.7 User Text Encoder

The original user text is processed using text encoder. The encoder converts the user's textual request into a feature representation. The same encoder weights are shared between the OCR text and user text branches so that the system can process language consistently regardless of its source.

3.1.8 Linear Projection

The outputs of the vision and text encoders may have different dimensions. Therefore, linear projection layers are used to transform the extracted features into a common feature dimension. In the proposed architecture, the features are projected into a shared dimension.

This common representation allows the visual features, OCR text features, and user text features to be combined effectively in the feature fusion module.

3.1.9 Classification Head

The fused multimodal representation is passed to a classification head. The classification head is a neural network that produces a jailbreak risk score representing the likelihood that the input contains a malicious or jailbreak attempt.

For example, a lower risk score represents a safer input, while a higher risk score indicates a potentially malicious input.

3.1.10 Feature Fusion

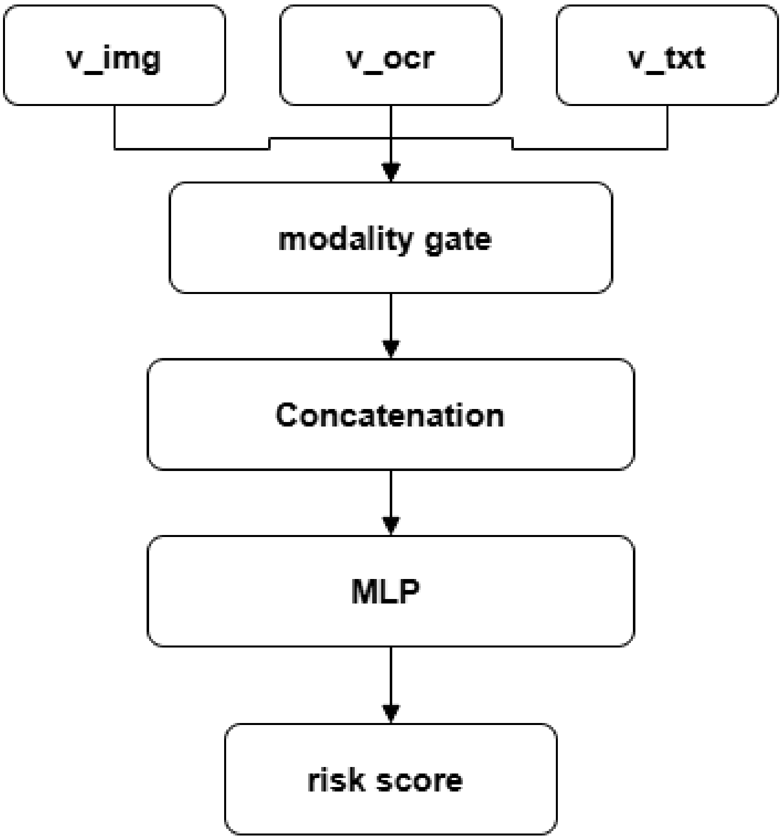

Figure 3.2: Feature Fusion Module Architecture

The baseline feature-fusion approach takes the feature representations from the image, OCR, and textual modalities as inputs [6]. The image representation captures visual information, the OCR representation captures text extracted from the image, and the textual representation captures information from the input prompt or request.

The three modality representations are first passed through a modality gating mechanism. The gate learns the relative importance of each modality and assigns appropriate weights to their corresponding feature representations. This allows the model to emphasize more informative modalities while reducing the influence of less relevant ones.

After the gating process, the resulting modality representations are concatenated to form a unified multimodal feature representation. This combined representation contains information from the image, OCR, and textual modalities and provides a common representation for subsequent processing.

The concatenated representation is then passed through a multilayer perceptron (MLP).

The MLP progressively transforms the fused representation into a compact feature representation while learning interactions among the different modalities. The resulting representation is finally used to produce the risk score of the input.

3.1.11 Calibrated Thresholds

The risk score is passed through calibrated decision thresholds. Two thresholds are used to divide the risk score into different decision regions. Based on the obtained risk score, the input is classified as *Allow* or *Block*.

3.1.12 Allow Decision

If the risk score is below the lower threshold, the input is classified as *Allow*. The request is considered sufficiently safe and is forwarded to the protected Vision-Language Model for further processing.

3.1.13 Block Decision

If the risk score is above the blocking threshold, the input is classified as *Block*. The request is considered potentially harmful or malicious and is rejected by the guardrail without being forwarded to the protected Vision-Language Model.

3.1.14 Protected Vision-Language Model

The protected VLM is the downstream model that processes the requests accepted by the guardrail. The guardrail acts as an additional security layer between the user and the VLM, preventing potentially malicious multimodal inputs from directly reaching the model.

CHAPTER 4

PROJECT WORK PLAN AND GANTT CHART

4.1 Project Execution Timeline

The project execution timeline spans a 5-month period from August 2026 to December 2026. The development milestones are structured across distinct technical phases, with project documentation and report writing conducted continuously from project initiation to final submission (Months M1 to M5).

4.2 Gantt Chart

The breakdown of tasks, execution periods, and development phases is illustrated in the Gantt Chart below:

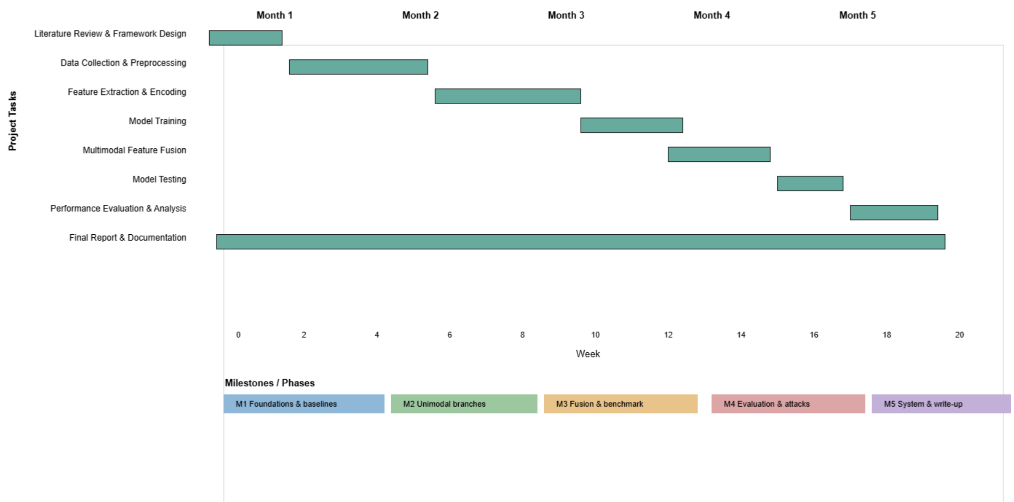

Figure 4.1: 5-Month Project Execution Gantt Chart (August 2026 – December 2026)

CHAPTER 5

CURRENT PROGRESS, DATASET COLLECTION, AND FUTURE WORK

5.1 Dataset Collection

A structured multimodal dataset has been collected to support the training and evaluation of the proposed guardrail. The dataset consists of paired text-and-image samples curated to represent both malicious (jailbreak) and benign interaction scenarios, ensuring that the model learns to distinguish genuinely harmful multimodal inputs from safe ones rather than relying on superficial patterns.

5.1.1 Data Categories

The collected dataset is organized into the following categories, aligned with the three primary jailbreak failure modes identified in Chapter 2:

- **Typographic Injection Samples:** Images in which harmful or policy-violating instructions are rendered directly as visual text (e.g., embedded within the image using overlays, stylized fonts, or scene-text placement). These samples are paired with minimal or neutral accompanying user text, isolating the image as the primary attack vector.
- **Query-Relevant Pairing Samples:** Samples where the text prompt and the image are individually benign but become harmful when interpreted together. These require the guardrail to reason jointly across modalities rather than evaluating text and image independently.
- **Adversarial Perturbation Samples:** Images subjected to small, optimized pixel-level perturbations designed to suppress the safety behavior of the downstream vision-language model, paired with otherwise ordinary or harmful text prompts. These samples test the guardrail's sensitivity to subtle visual manipulations that are not perceptible to human reviewers.
- **Benign Samples:** A diverse set of safe, policy-compliant text-and-image pairs

spanning everyday queries, general knowledge questions, and neutral image content. These are essential for calibrating the guardrail's thresholds and reducing false-positive rates.

5.1.2 Custom Dataset Curation

In addition to leveraging existing labeled samples, a custom data collection process was carried out to increase category coverage and reduce dependency on a single data distribution. This involved:

- Manually curating additional query-relevant pairing examples by combining benign images with contextually related text prompts to simulate realistic attack scenarios not well represented in existing samples.
- Generating supplementary typographic injection samples by rendering varied instruction text onto diverse image backgrounds, fonts, sizes, and orientations to improve OCR-branch robustness.
- Collecting additional benign multimodal pairs from general-purpose image-text sources to balance the ratio of safe to unsafe samples and reduce class imbalance during training.
- Applying a manual review and relabeling pass on a subset of the aggregated dataset to verify category correctness and remove ambiguous or mislabeled samples.

5.1.3 Dataset Summary

Table 5.1 presents a high-level summary of the dataset composition across the defined categories.

Table 5.1: Summary of collected dataset by category.

Category	**Modality**	**Purpose**
Typographic Injection	Image + Text	Test OCR-branch attack detection
Query-Relevant Pairing	Image + Text	Test cross-modal reasoning
Adversarial Perturbation	Image + Text	Test robustness to pixel-level noise
Benign	Image + Text	Threshold calibration, false-positive control

The dataset is currently being preprocessed and split into training, validation, and test partitions, with stratified sampling used to preserve category proportions across splits.

5.2 Current Progress

The project has progressed through data preparation and has moved into system-facing development. Key achievements are summarized below:

- **Dataset Collection and Curation:** A categorized multimodal dataset covering typographic injection, query-relevant pairing, adversarial perturbation, and benign samples has been collected, supplemented with custom-curated data, and is undergoing preprocessing.
- **System Architecture Finalization:** The tri-branch encoder design, feature fusion module, and calibrated threshold-based decision mechanism have been finalized, as detailed in Chapter 3.
- **User Interface Development:** An interactive user interface is under active development to allow end users to submit multimodal requests and view the guardrail's classification decision in real time.

5.2.1 User Interface Design and Placement

The user interface serves as the front-facing interaction layer of the system, allowing a user to upload an image, enter an accompanying text prompt, and submit the combined request to the guardrail pipeline. Upon submission, the interface displays the computed risk score along with the final decision (Allow or Block). If the request is allowed, it is forwarded to the protected Vision-Language Model and the generated response is displayed to the user; if blocked, a rejection notice is shown instead.

Figure **??** illustrates the placement of the user interface within the overall system workflow, positioned as the entry point through which all user requests are submitted before reaching the input router described in Section 3.1.2.

As shown in Figure **??**, the interface consists of three main regions: (i) an image upload panel, (ii) a text input field for the accompanying prompt, and (iii) a results panel displaying the risk score, decision label, and, when applicable, the downstream model's response.

5.3 Challenges

Several challenges have been encountered and are anticipated during the remaining phases of the project:

- **Data Imbalance:** Jailbreak samples, particularly adversarial perturbation examples, are comparatively harder to source and construct than benign samples, leading to class imbalance during model training. Careful resampling and augmentation strategies are being explored to mitigate this.
- **OCR Reliability:** Optical Character Recognition accuracy can degrade on stylized, distorted, low-contrast, or unusually oriented typographic injections, potentially reducing the effectiveness of the OCR text branch and requiring additional preprocessing steps.
- **Label Ambiguity:** Some collected samples, particularly in the query-relevant pairing category, require subjective judgment to label correctly, introducing potential inconsistency during manual review.
- **Generalization to Unseen Attacks:** Since jailbreak techniques evolve rapidly, ensuring the guardrail generalizes to attack variants not present in the training data remains a significant challenge.
- **Computational Constraints:** Balancing detection accuracy with the lightweight, low-latency design goal of the guardrail requires careful tuning of encoder sizes, projection dimensions, and fusion module complexity.
- **UI-Backend Integration:** Ensuring smooth, low-latency communication between the user interface and the backend guardrail pipeline, particularly for image uploads and real-time score display, requires further engineering effort.

5.4 Future Work

The following tasks are planned for the upcoming phases of the project:

- Complete preprocessing, cleaning, and augmentation of the collected dataset to address class imbalance and improve robustness across all four categories.

- Implement and train the vision encoder, OCR text encoder, and user text encoder branches, followed by integration of the feature fusion module and classification head.
- Finalize the user interface and complete its integration with the backend guardrail pipeline for end-to-end testing.
- Conduct threshold calibration experiments to optimize the Allow/Block decision boundary and minimize false-positive rates.
- Perform ablation studies to evaluate the individual contribution of each branch (raw image, OCR text, user text) to overall detection performance.
- Evaluate the complete system using accuracy, precision, recall, F1-score, and false-positive rate, and compare results against existing baseline guardrail approaches discussed in Chapter 2.
- Conduct user testing on the interface to gather feedback on usability and refine the results display accordingly.

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